

Lab: Install Helm (Kubernetes Package Manager)

What is Helm?

Helm is a **package manager for Kubernetes**.

It helps you **install, upgrade, rollback, and manage applications** using **Helm charts**.

Pre-requisites (Mandatory)

Before installing Helm, ensure:

kubectl version --client

kubectl get nodes

Kubernetes cluster is reachable

kubectl is installed and configured

Helm **does NOT** require Tiller (removed since Helm v3)

Option 1: Install Helm on Linux (Ubuntu / Debian)

Step 1: Download Helm installation script

```
curl -fsSL https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
```

Step 2: Verify installation

```
helm version
```

Expected output:

```
version.BuildInfo{Version:"v3.x.x", ...}
```

Option 2: Install Helm on Rocky Linux / RHEL / CentOS

Step 1: Download Helm binary

```
curl -LO https://get.helm.sh/helm-v3.14.4-linux-amd64.tar.gz
```

Step 2: Extract and move binary

```
tar -xvf helm-v3.14.4-linux-amd64.tar.gz
```

```
sudo mv linux-amd64/helm /usr/local/bin/helm
```

Step 3: Verify

helm version

Option 3: Install Helm on macOS

Using Homebrew (Recommended)

brew install helm

Verify:

helm version

Option 4: Install Helm on Windows

Method 1: Using Chocolatey

choco install kubernetes-helm

Method 2: Manual

1. Download from:
<https://github.com/helm/helm/releases>
2. Extract helm.exe
3. Add it to **System PATH**

Verify:

helm version

Step 5: Validate Helm with Kubernetes

Check Helm environment

helm env

Check cluster access

helm list

If no error → Helm is correctly configured

Step 6: Add Official Helm Repository (Recommended)

helm repo add bitnami <https://charts.bitnami.com/bitnami>

helm repo update

Verify:

helm repo list

Step 7: Test Helm with a Sample Chart

Install Nginx using Helm

helm install test-nginx bitnami/nginx

Verify:

helm list

kubectl get pods

kubectl get svc

Step 8: Uninstall Test Release

helm uninstall test-nginx

Common Helm Commands (Students MUST remember)

Command	Purpose
helm create mychart	Create new chart
helm lint	Validate chart
helm template	Render YAML
helm install	Install chart
helm upgrade	Update release
helm rollback	Rollback version

Command	Purpose
helm uninstall	Remove release

Troubleshooting

helm: command not found

➔ Check PATH:

which helm

echo \$PATH

Kubernetes cluster unreachable

➔ Fix kubeconfig:

kubectrl config view

kubectrl get nodes